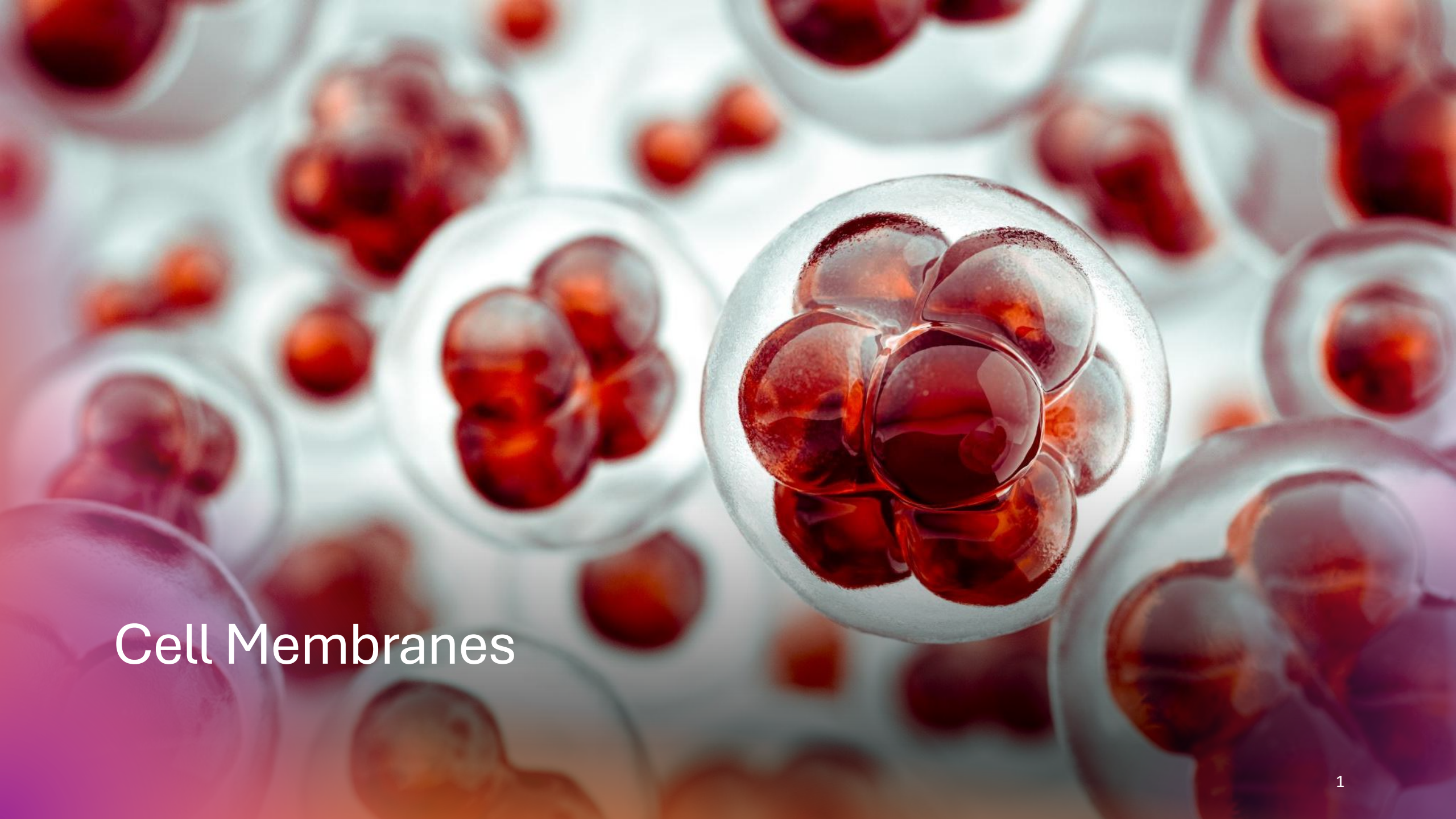
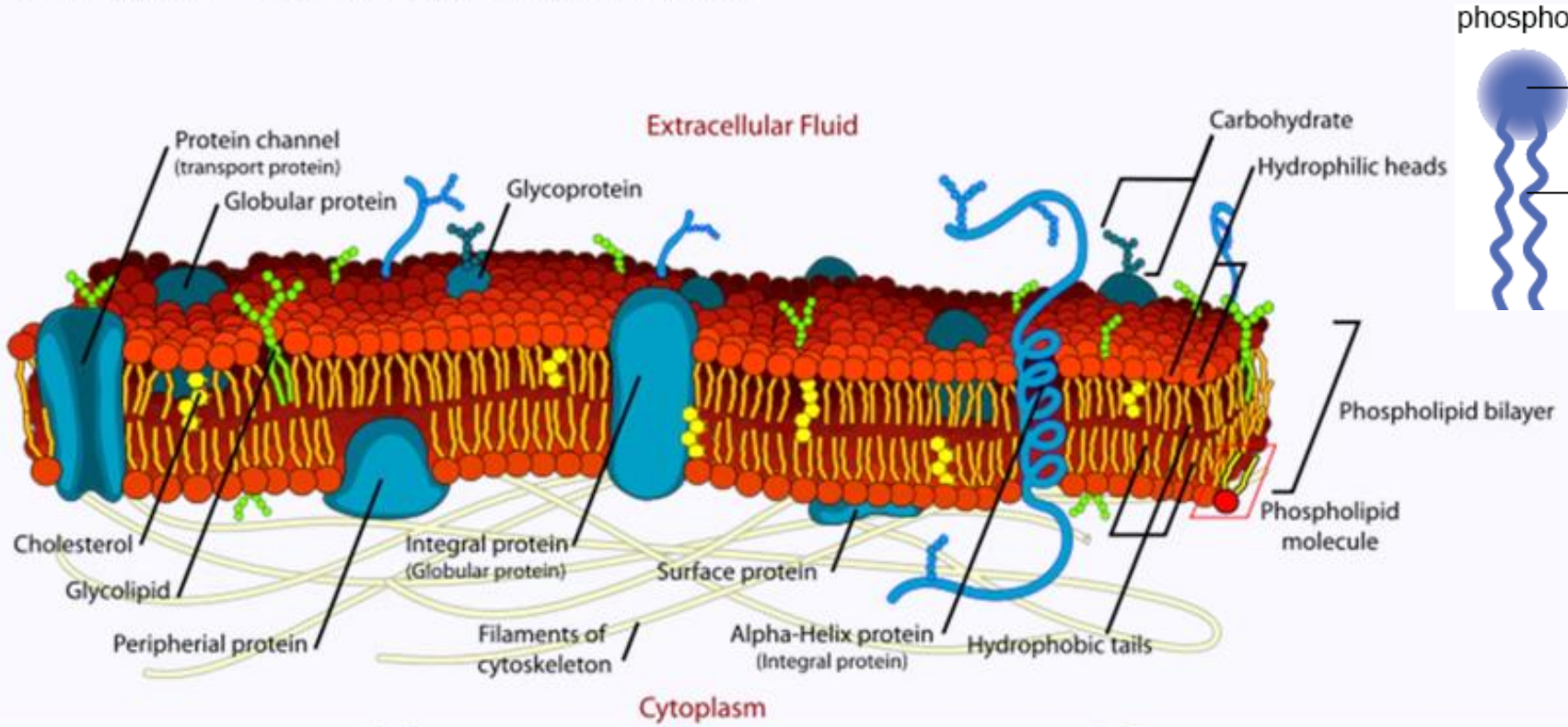
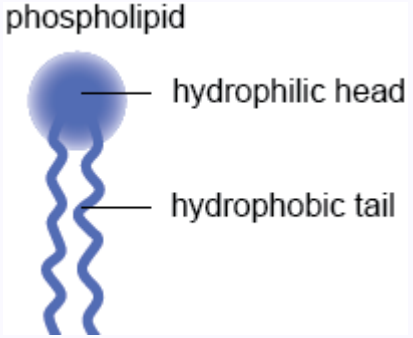


Cell Membranes

Structure of the Cell Membrane





Phospholipid Bilayer

Two layers of phospholipids form the foundation, with hydrophilic heads facing outward and hydrophobic tails facing inward.

Membrane Proteins

Proteins embedded throughout the membrane perform specialized functions like transport and cell signaling.

Fluid Mosaic Model

The membrane is fluid and dynamic, with proteins floating like islands in a sea of lipids.

Functions of the Cell Membrane

01

Mechanical Structure

Provides shape and protection, maintaining cell integrity against external forces.

02

Selective Permeability

Acts as a gatekeeper, allowing only specific substances to pass through.

03

Transport

Facilitates movement of essential molecules into and out of the cell.

04

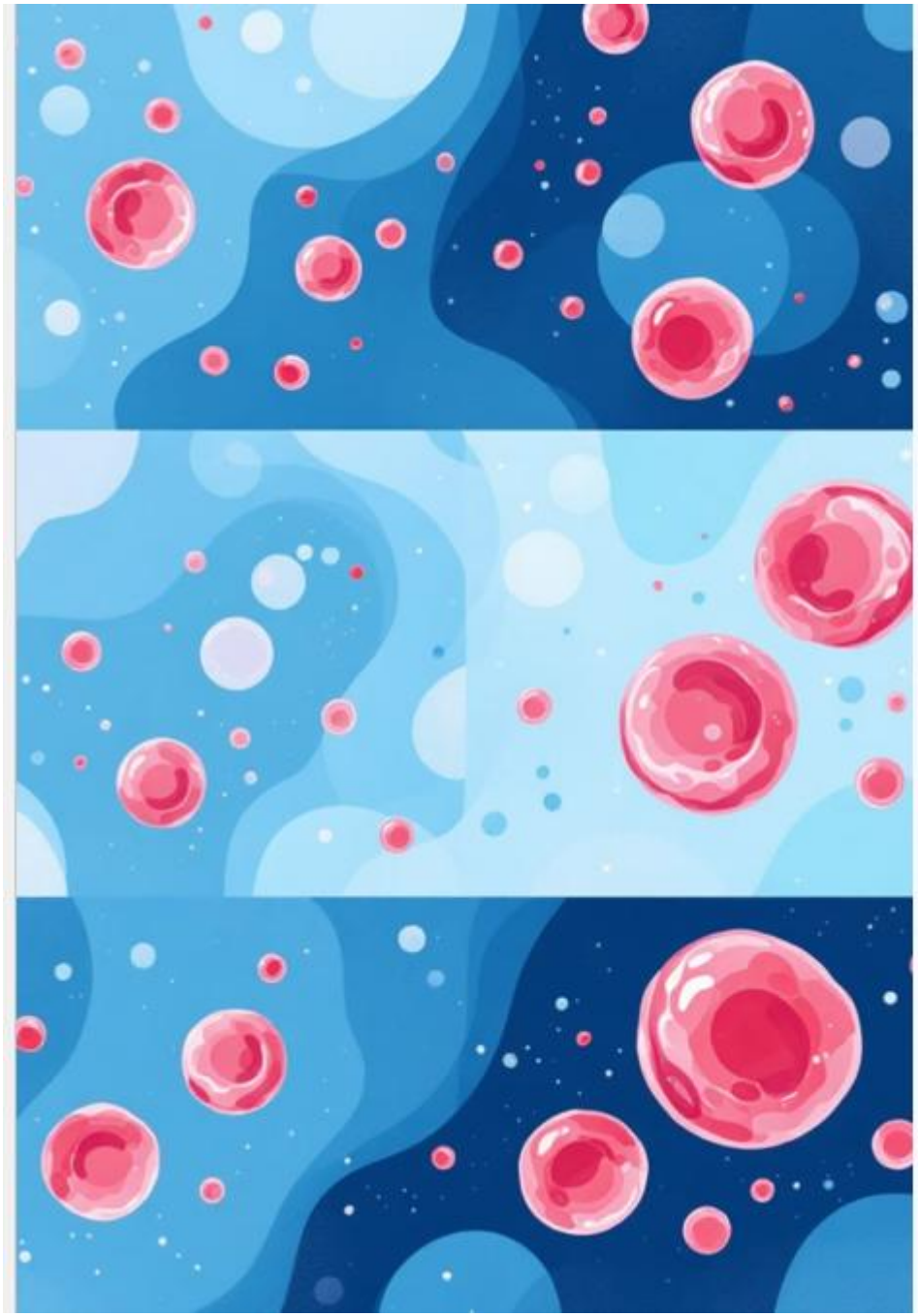
Cell Signaling

Contains receptors that recognize chemical signals from other cells.

05

Metabolic Activities

Hosts enzymes that catalyze important biochemical reactions at the membrane surface.



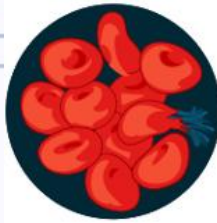
Solution Concentrations

The movement of water across membranes depends on the relative concentration of solutes on either side.



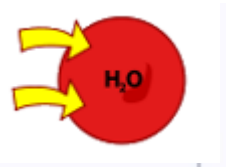
Isotonic Solution

Solute concentration is equal inside and outside the cell. Water moves equally in both directions, resulting in no net change in cell size.



Hypotonic Solution

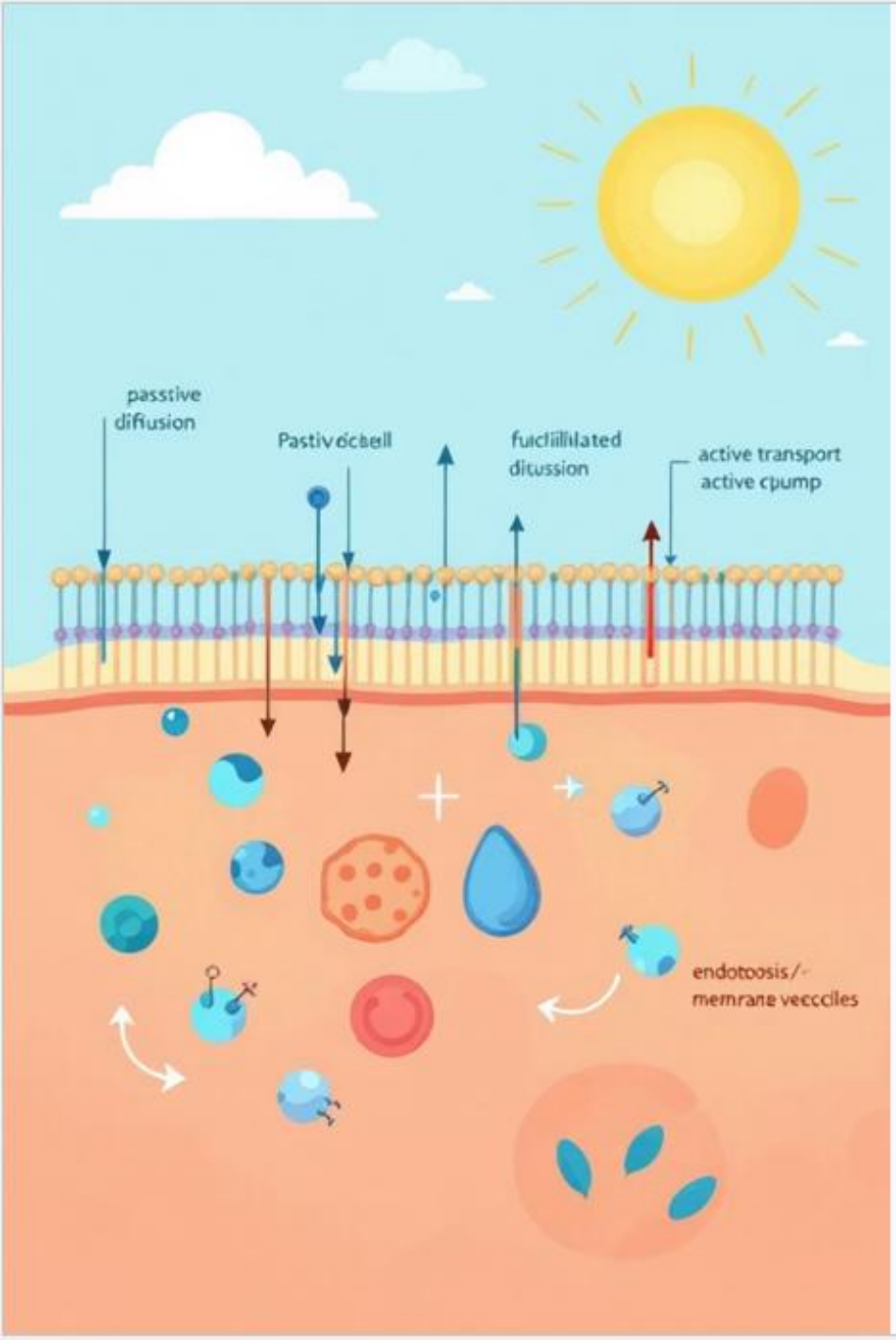
Lower solute concentration outside the cell. Water moves into the cell, causing it to swell or potentially burst (lysis).



Hypertonic Solution

Higher solute concentration outside the cell. Water moves out of the cell, causing it to shrink (crenation).





Transport Across the Cell Membrane

1

Passive Transport

No energy required. Materials move from high to low concentration.

2

Active Transport

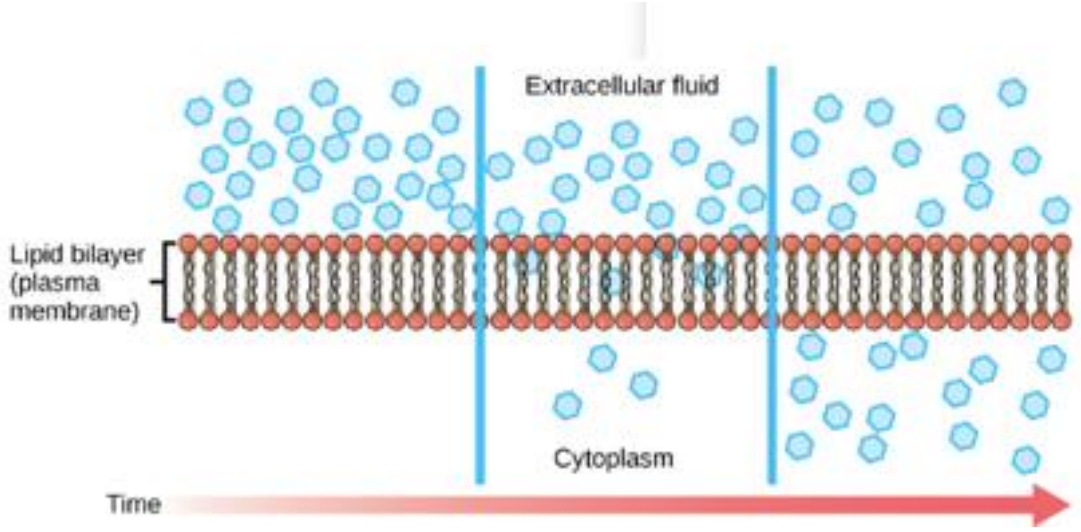
Energy (ATP) required. Materials move from low to high concentration.

Passive Transport

Net movement from high concentration to low concentration.

- ✓ Passive biological process.
- ✓ Zero cellular energy required.
- ✓ Driven by kinetic energy.

The Drive to Equilibrium





Simple Diffusion

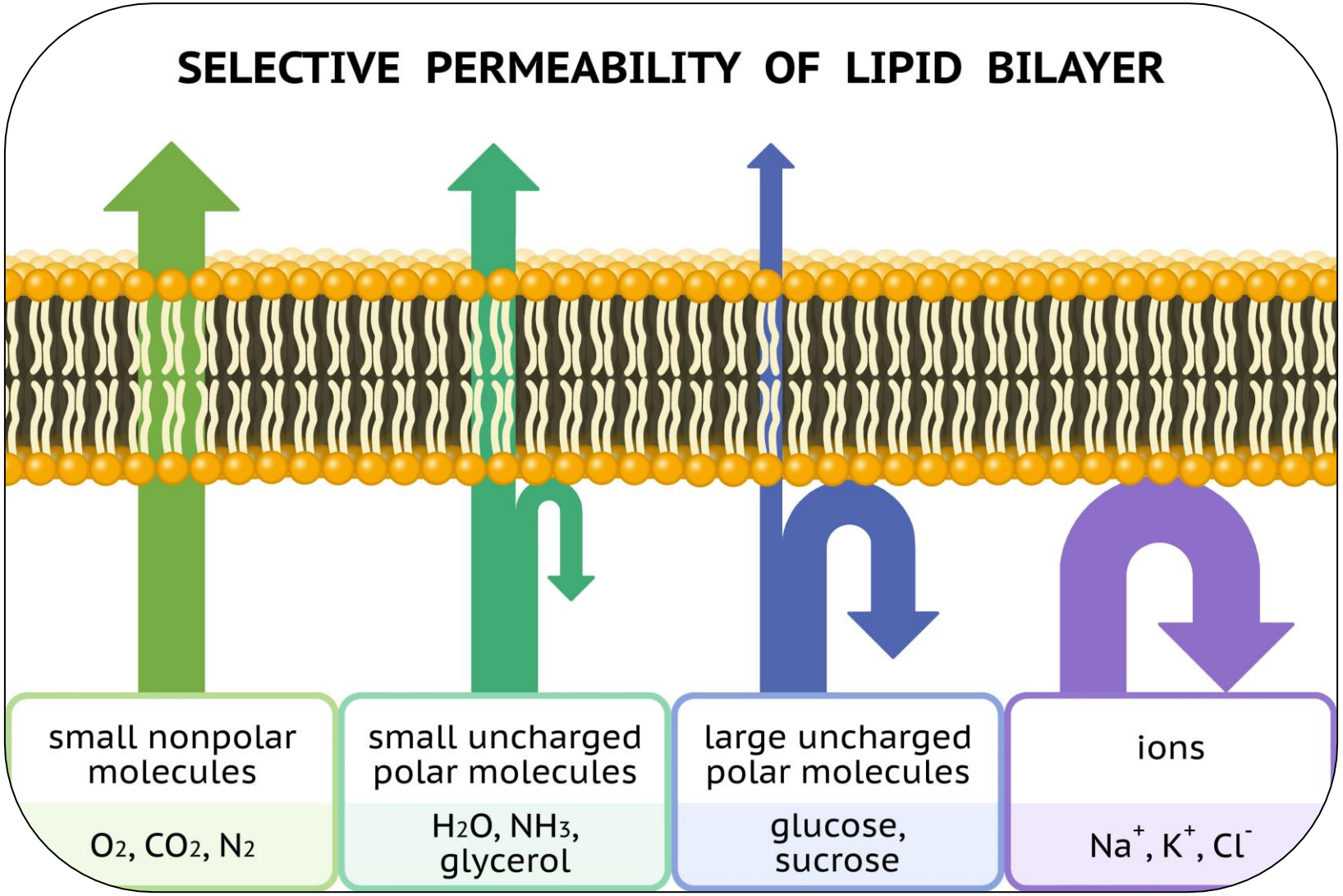
Direct passage through the phospholipid bilayer without structural assistance.



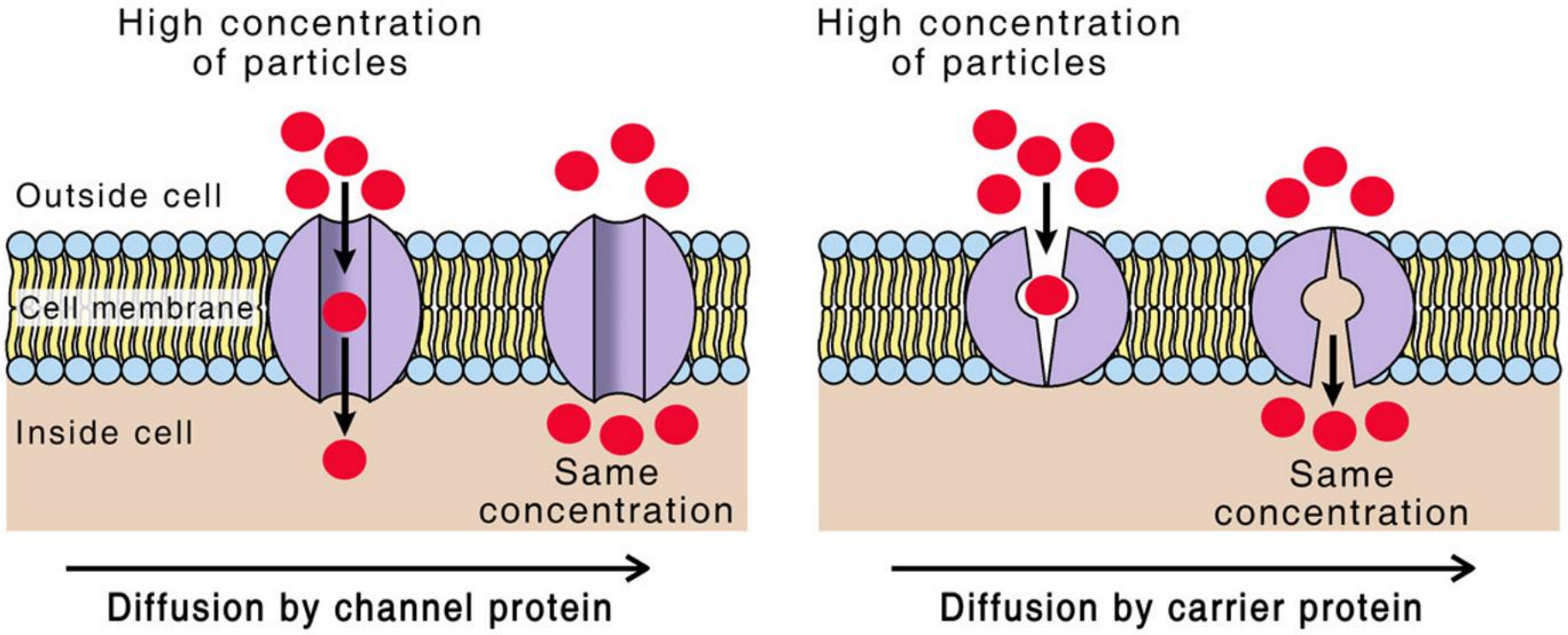
Facilitated Diffusion

Mediated passage across the membrane utilizing specific transport proteins.

Simple Diffusion



Facilities Diffusion



Active Transport

Defying Equilibrium

Net movement from low concentration to high concentration.

- Moves against gradient.
- Requires cellular energy (ATP).
- Utilizes integral carrier proteins.

Classification of Active Transport



Primary (Direct)

Directly hydrolyzes ATP to translocate solutes against their gradient.



Secondary (Coupled)

Exploits an existing electrochemical gradient to drive transport.

PRIMARY ACTIVE TRANSPORT

Essential Ion Pumps



Na^+/K^+-ATPase

Maintains resting membrane potential.



Ca^{2+}-ATPase

Muscle contraction regulation.



H^+/K^+-ATPase

Gastric acid secretion in stomach.

SECONDARY ACTIVE TRANSPORT

Energy Source

Pre-existing electrochemical gradient.

Dependency

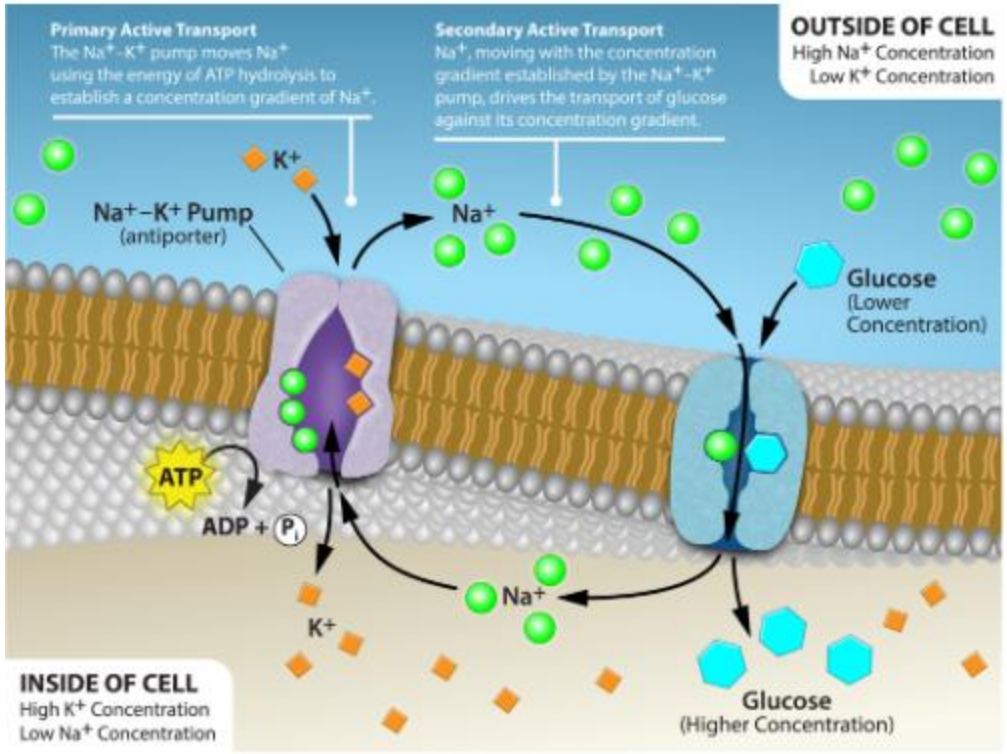
Relies entirely on primary pumps.

Transporter Type

Symporters and antiporters.

Mechanism

Couples downhill ion flow with uphill cargo.



SECONDARY ACTIVE TRANSPORT

Mechanisms of Coupling

Symport (Cotransport)

Driving ion and cargo move in the same direction. (e.g., Na^+-glucose symporter, SGLT1).

Antiport (Exchange)

Driving ion and cargo move in opposite directions. (e.g., Na^+/Ca^{2+} exchanger, NCX).

Endocytosis and Exocytosis



Endocytosis

The cell membrane wraps around external material, forming a vesicle that brings substances into the cell. This process requires energy.



Exocytosis

Vesicles inside the cell fuse with the membrane, releasing their contents to the external environment. Used for waste removal and secretion.



Types of Endocytosis

Phagocytosis

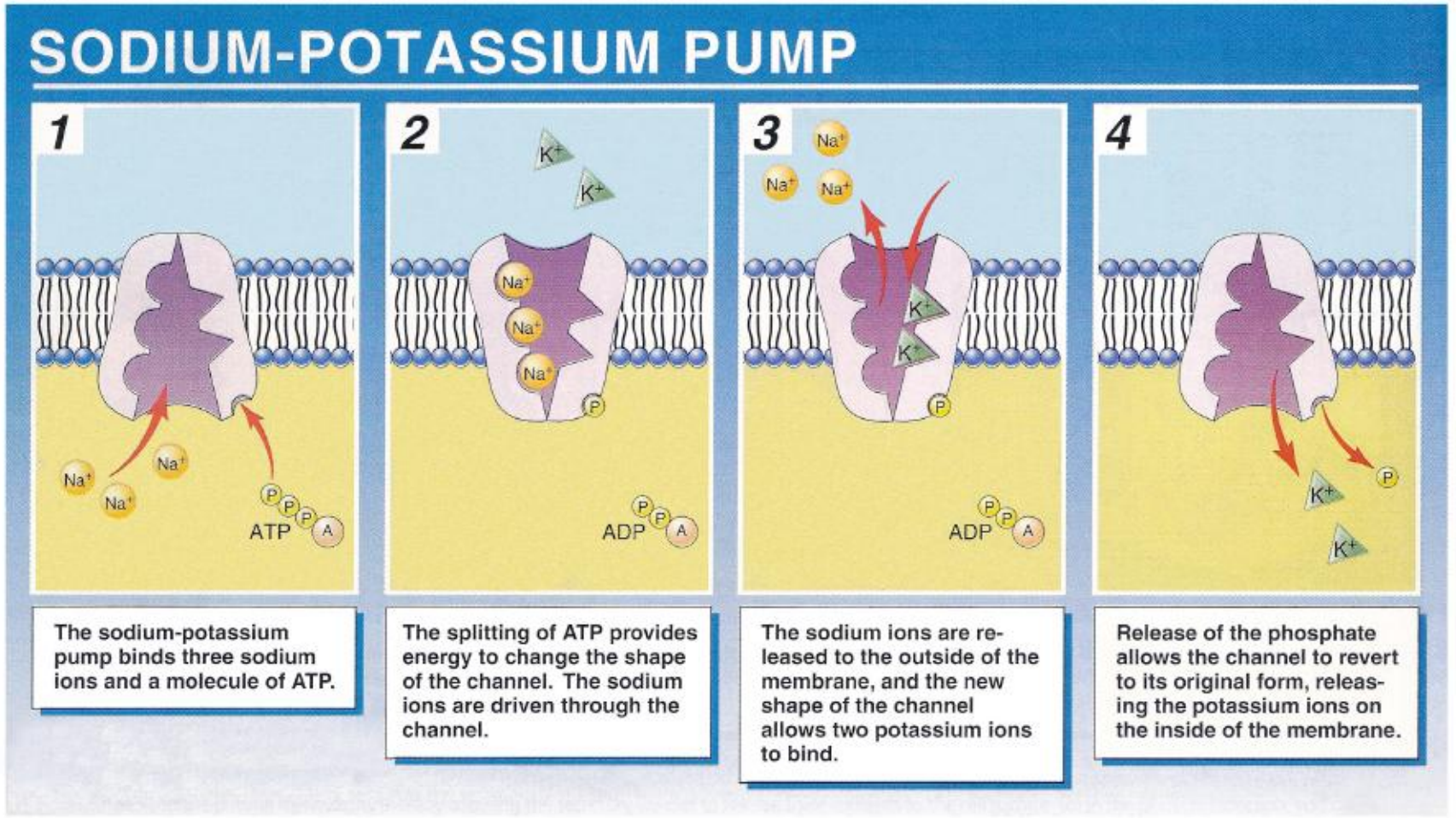
"Cell eating" - the cell engulfs large particles or entire cells. Used by immune cells to capture pathogens.

Pinocytosis

"Cell drinking" - the cell takes in dissolved substances and fluids. A continuous process for nutrient absorption.

Receptor-Mediated

Specific molecules bind to receptors on the membrane, triggering vesicle formation. Highly selective and efficient.



Sodium-Potassium pump

Cell membrane transport review

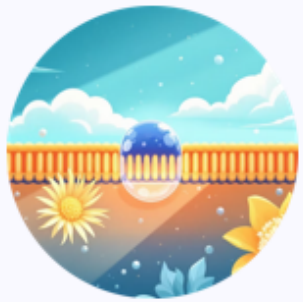


Cell Membrane Transport Review



Simple Diffusion

Small, nonpolar molecules move directly through the membrane from high to low concentration without assistance.



Facilitated Diffusion

Channel or carrier proteins help polar or charged molecules cross the membrane without energy input.



Active Transport

Protein pumps use ATP energy to move molecules against their concentration gradient from low to high.



Bulk Transport

Large particles and macromolecules move via endocytosis (into cell) or exocytosis (out of cell) using vesicles.